



Motilal Oswal
Centre for
Capital Markets

MOCCM Quant Hackathon 2026: The Black-Box Signal Hunt

Section A: Participant Briefing & Rulebook

Hosted by: Motilal Oswal Centre for Capital Markets (MOCCM), IIT Bombay in collaboration with Quant Club and WnCC, IIT Bombay.

Duration: 48-Hour Sprint **Location:** Local Compute / Virtual

Judges

- Jatin Phutela, Chief Business Officer and Global CFO, Graviton Research
- Manish Mahajan, Founder, Cloudcraftz (Singapore)
- Rajib Ranjan Borah, Founder, iRage
- Debajyoti Nandy, Quant analyst, IMC Trading

Key times

10:45AM Friday 5th June: Virtual kick-off of the hackathon – Amit Chakarabarty (MOCCM) and Debajyoti Nandy (IMC Trading) will be present.

11AM Friday 5th June: 10 year training dataset released – problem statement is already outlined in this document. Teams have 48 hours (till 11AM on 7th June) to submit their final code and presentation.

11AM Saturday 6th June: “Ask me anything” (AMA) session – to clarify any doubts participants might have – Amit Chakarabarty (MOCCM) and Debajyoti Nandy (IMC Trading) will be present.

11AM Sunday 7th June: Code freeze. Submit your code and your presentation in the GitHub repository. Follow the naming conventions outlined below, otherwise the submissions might be disqualified.

11:15AM Sunday 7th June: New 5 year dataset released – use this dataset to run your code and generate the output files (which will be evaluated). The two files should be named ***teamname_longonly_results.csv*** and ***teamname_longshort_results.csv*** (follow the naming convention).

11:30AM Sunday 7th June: The two files ***teamname_longonly_results.csv*** and ***teamname_longshort_results.csv*** should be uploaded on the Google form provided. Please note that Google form will be **closed at 11:30AM sharp**.

12:15PM Sunday 7th June: The results will be announced and the top 5 teams will be asked to present in front of the panel of judges starting 12:30PM. Draw of lots to decide who goes first, second, and so on.

12:30PM Sunday 7th June: Presentations start. Each time has 10 mins for presentation and 8 mins for Q&A.

2PM Sunday 7th June: Judges and organizers deliberate and the top 3 winners identified.

2:15PM Sunday 7th June: Announce 1st, 2nd and 3rd prize winners. Top 5 teams will also receive certificates at a later date.



1. Objective

Welcome to the Black-Box Signal Hunt. In standard quantitative finance, analysts search for "Alpha" – a trading signal that generates consistent, risk-adjusted returns.

For this hackathon, we have synthesized a 10-year market dataset containing 50 assets. Hidden deep within this data is a complex, multi-dimensional, deterministic trading signal. Your objective is to mathematically isolate this signal and build a custom, ultra-low-latency backtesting engine to prove your strategy works.

2. Dataset Schema

At the exact start time of the hackathon, all registered teams will receive a link to download **moccm_intraday_blackbox.csv** (500MB file). To thoroughly test your system architecture, the dataset contains roughly 9.45 million rows of 5-minute Intraday data formatted as follows:

Date	Ticker	Close	Volume
2016-01-01 09:15:00	TICKER_00	145.22	1204500
2016-01-01 09:15:00	TICKER_01	89.40	3400000

- Target Asset: You are only allowed to trade TICKER_00.
- Predictors: The other 49 tickers contain the variables, conditions, and triggers that dictate the future price movements of TICKER_00.

3. The Dual Challenge

To win, your team must successfully execute on two fronts:

A. Quant Challenge (Signal Discovery) You must find the hidden rule. It is not a simple moving average crossover. The signal is non-linear and conditional, relying on cross-asset correlations, volume thresholds, and specific mathematical states. Furthermore, your strategies must survive a strict 10 basis points (0.10%) transaction fee per trade.

B. Engineering Challenge (Systems & Latency) Standard Python libraries like pandas or Backtrader are slow. You must build a custom backtesting engine (in C++, Rust, or heavily vectorized NumPy) capable of parsing large CSV files and running a full simulation in under 30 seconds.

4. Rules & "White-Box" Mandate

This competition is designed to test human intuition and raw system architecture, not your ability to call an API.

- **Rule 1: NO Machine Learning / AI.** The use of Neural Networks, Random Forests, LLMs, or libraries like PyTorch, TensorFlow, or scikit-learn is strictly prohibited.
- **Rule 2: White-Box Logic Only.** You must be able to explicitly write out the mathematical logic of your signal (e.g., "If Asset X does this, and Asset Y does this, buy Asset Z"). Black-box AI models that cannot explain their reasoning will be disqualified.



- **Rule 3: Local Compute Only.** You may not use cloud computing clusters. Your backtesting engine must run locally on your team's laptops.
- **Rule 4: Capital Limits & Mandatory Strategies.** You must submit TWO distinct strategies based on your discovered signal:
 1. Long-Only strategy (capped at strictly \$1,000,000 gross exposure).
 2. Long-Short strategy (capped at strictly \$2,000,000 absolute gross exposure).

5. Mandatory Output Schema (Audit Log)

To verify your strategy complies with the capital limits, margin requirements, and transaction friction, your custom backtesting engine must output a standardized audit log.

For your Long-Only and Long-Short submissions, your engine must generate two CSV files named ***teamname_longonly_results.csv*** and ***teamname_longshort_results.csv*** respectively (so two .csv files in total, teamname in lowercase).

Each CSV must contain a row for every 5-minute interval in the dataset with the following exact columns:

1. Timestamp: The 5-minute interval time.
2. Gross_Exposure: The absolute total dollar value of all active positions (Longs + Shorts) during this interval.
3. Cash_Balance: Uninvested cash remaining. *(If this drops below 0, you are disqualified for a margin call).*
4. Interval_Turnover: The total dollar amount of assets traded (bought + sold) during this specific 5-minute window. *(MOCCM will automatically deduct 10 bps from this number to calculate your transaction fees).*
5. Gross_NAV: The total portfolio value (Cash + Active Positions) *before* fees are applied.

Note: The judges will run an automated script on your CSVs to deduct the 10 bps fee from your Interval_Turnover, calculate your True Net Asset Value (NAV), and determine your Blended Net Sharpe Ratio. If your engine fails to output this exact format, your submission cannot be graded.

Script that generates a dummy output file (to cross-check the engine's csv output format is accurate)

```
import pandas as pd
```

```
import numpy as np
```

```
from datetime import datetime, timedelta
```

```
def generate_test_sample(filename="test_sample_output.csv"):
```

```
    """
```



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```
Generates a 100-row dummy file to test if your engine  
matches the MOCCM Grader schema.  
""""  
  
print(f"Generating sample output: {filename}")  
  
  
# Create 100 intervals of 5-minute data  
start_time = datetime(2026, 1, 1, 9, 15)  
timestamps = [start_time + timedelta(minutes=5*i) for i in range(100)]  
  
# Generate dummy data  
data = {  
    'Timestamp': timestamps,  
    'Gross_Exposure': np.random.uniform(500000, 900000, 100),  
    'Cash_Balance': np.random.uniform(100000, 500000, 100),  
    'Interval_Turnover': np.random.uniform(5000, 20000, 100),  
    'Gross_NAV': np.random.uniform(1000000, 1100000, 100)  
}  
  
df = pd.DataFrame(data)  
df.to_csv(filename, index=False)  
  
print("Test file generated. Ensure your engine produces a CSV identical to this  
format.")  
  
if __name__ == "__main__":  
    generate_test_sample()
```



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6. Finale: The 15-Minute "Drop"

We are not judging you on the 10-year training dataset. We are judging you on how your engine handles the unknown.

1. Sunday at 11:00 AM: Coding stops. You must commit your backtesting engine code (***teamname_code*** – **C++, Rust or Python/NumPy**) and your mathematical strategy (***teamname_proof.pdf***) to your team's GitHub repository.
2. Sunday at 11:15 AM (The Drop): MOCCM will release a brand new, hidden 5-year dataset.
3. The Race: You have exactly 15 minutes to download the new dataset, run it through your pre-built backtesting engine, and submit your final two CSV files named ***teamname_longonly_results.csv*** and ***teamname_longshort_results.csv*** on this new dataset.

Note: If your engine is slow, you will miss the 15-minute submission window and score a zero. If you overfit a model to the training data instead of finding the true mathematical signal, your P&L will collapse on the new dataset.

Section B: Evaluation Rubric

Phase 1: Automated Screening (15-Minute Drop)

If an engine trips any of the below rules on the hidden 5-year data, it scores zero:

- Speed Test: The full backtest runs in less than 30 seconds.
- Capital Ceiling Limit: Never exceeds \$1 million (Long-Only) or \$2 million (Long-Short) in gross exposure.
- Margin Check: Cash balance never falls below 0.

Quantitative Scoring: Submissions will be ranked purely by their Blended Net Sharpe Ratio (the average Sharpe Ratio of their Long-Only and Long-Short submissions) on the hidden 5-year dataset. The Net Sharpe Ratio is calculated *after* deducting 10 bps for every transaction. Maximum Drawdown (MDD) and Return on Deployed Capital might be used as secondary ranking metrics.

Phase 2: Live Jury Defense (100 Points)

The Top 5 most profitable teams will present their mathematical proofs and system architecture to a judges' panel.

- Signal Discovery & Math Proof (30 Pts): Can the team write out their "White-Box" math logic clearly and prove how they discovered the hidden rule? Does the logic rely only on data to execute trades?
- Strategy Architecture & Constraints (30 Pts): How did the team adjust their rebalancing frequency to survive the 10 bps fee? How did they manage position sizing between their two mandatory strategies?
- Dual Portfolio Defense (20 Pts): Since both portfolios are required, how did the Long-Only strategy protect against drawdowns without shorting? Conversely, did the Long-Short strategy actively use short positions to neutralize broader market risk (Beta), or did they just double their leverage recklessly?



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- Engineering & Scalability (20 Pts): What languages/libraries were used to guarantee sub-30-second execution on a large dataset? How did they manage RAM?

This will be a LIVE ~18 mins presentation by the Top 5 teams, structured as follows (subject to change):

- Pitch (10 Minutes): Team presents their methodology, their system architecture, and their mathematical proof.
- Q&A (8 Minutes): Questions from the judges.